MIL-STD-878A
22 October 1969
Superseding
MIL-STD-878
17 August 1967

MILITARY STANDARD

# METHOD OF DIMENSIONING AND DETERMINING CLEARANCE FOR AIRCRAFT TIRES AND RIMS

FSC 2620

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
WASHINGTON, D. C. 20301

Method of Dimensioning and Determining Clearance for Aircraft Tires and Rims

MIL-STD-878A

1. This Military Standard is mandatory for use by all Departments and Agencies of the Department of Defense.

*2. Recommended corrections, additions, or deletions should be addressed to the Aeronautical Systems Division, Attn: ASNPS-30, Wright-Patterson Air Force Base, Ohio 4543[illegible].



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.

## * CONTENTS


| | | | Page |
|---|---|---|---|
| Paragraph | 1 | SCOPE | 1 |
| | 1.1 | Purpose | 1 |
| | 2 | REFERENCED DOCUMENTS | 1 |
| | 3 | GENERAL REQUIREMENTS | 1 |
| | 3.1 | Proportions and clearances | 1 |
| | 3.2 | Definitions of figure 2 | 2 |
| | 4 | INTERNATIONAL STANDARDIZATION | 4 |
| | 4.1 | International standardization agreement | 4 |


### FIGURES


| | | | |
|---|---|---|---|
| FIGURE | 1. | Profile Proportions for Aircraft Landing Wheel Tires | 5 |
| | 2. | Chart for Obtaining Radial Clearance $C_R$ and Lateral Clearance $C_W$ | 6 |
| | 3. | Design Guide for Obtaining Dimension "A" | 7 |
| | 4. | Aircraft Wheel Rim Proportions | 8 |


### TABLES


| | | | |
|---|---|---|---|
| TABLE | I. | Growth factors | 3 |




Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.

# METHOD OF DIMENSIONING AND DETERMINING CLEARANCE FOR AIRCRAFT TIRES AND RIMS

## 1. SCOPE

1.1 Purpose. This standard establishes the procedures for dimensioning and determining clearance for aircraft tires and rims.

## 2. REFERENCED DOCUMENTS

*2.1 The issue of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-T-5041 | Tires, Pneumatic, Aircraft |
| MIL-R-7726 | Repair and Retreading of Used Pneumatic Tires and Repair of Inner Tubes |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. GENERAL REQUIREMENTS

3.1 Proportions and clearances. The proportions and clearances of aircraft wheels and tires shall be in accordance with figure 1. Figure 1 is based upon the following considerations:

(a) The dimensions of new, unused inflated tires after a 12-hour inflation period at rated pressure to remove initial tire stretch shall not exceed dimensions $D_O$, H, W, $D_S$, and $W_S$.

(b) Provisions shall be made for tire growth represented by dimensions $W_G$, $D_G$, $W_{SG}$, and $D_{SG}$. Tire growth is based upon the factors listed in table I.

(c) Minimum radial and lateral clearances beyond tire growth limits are derived from figure 2.

(d) Tire aspect ratios are obtained from figure 3.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.

*(e) The maximum shoulder dimensions shown as $H_S$ and $W_S$ shall be 0.82 of the maximum section height for all tire types, 0.85 of the maximum section width for type III tires, and 0.88 of the maximum section width for types VII and VIII tires. Types III, VII, and VIII tires are described in MIL-T-5041 and MIL-R-7726.

(f) Radii $\frac{W_S}{2}$ and $\frac{W_{SG}}{2}$ are drawn through their respective shoulder points tangentially to $D_O$ and $D_G$, respectively. Radii below the shoulder points pass through the shoulder points and are tangential to W and $W_G$, respectively.

(g) Dimension A, the wheel width between flanges, shall be obtained from figure 3.

3.2 Definitions of figure 2. Clearance allowance between the tire and the adjacent parts of the aircraft should be based on the maximum overall tire dimensions, plus growth allowance due to service, plus the increase in diameter due to centrifugal force. Minimum distances to adjacent parts of the aircraft should be determined as follows:

(a) Determine maximum grown tire envelope (i.e., the dotted line labeled "Grown Inflated Tire" on figure 1).

(b) Obtain radial clearance $C_R$ and lateral clearance $C_W$ from the chart on figure 2.

(c) Determine distance to adjacent part as follows:

Rx (min) = Radial distance from axle centerline (CL) to adjacent part = $\frac{D_G}{2} + C_R$

Wx (min) = Lateral distance from the CL to adjacent part = $\frac{W_G}{2} + C_W$

Sx (min) = Clearance allowed between tire shoulder area and adjacent part = $\frac{C_W + C_R}{2}$



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.

* TABLE I. Growth factors

| Classification | | Growth factor | |
|---|---|---|---|
| T and RA type | Squatness ratio | $G_W$ | $G_H$ |
| T and RA (1960) | No recommended limits | 1.03 | 2.06 |
| Type III | No recommended limits | 1.04 | 2.08 |
| Type VII | 0.78 and upwards | 1.05 | 2.12 |
| * Type VIII | 0.65 to 0.77 | 1.05 | 2.14 |

NOTES ON TABLE I:

*1. The general squatness ratios, aspect ratio $\frac{H}{W}$ of figure 4, i.e., mean section height/mean section width, are shown in table I but should be used as a design guide only.

2. For new design, the new inflated tire dimensions W, H, $W_S$, and $H_S$ are factorized with the appropriate factors for T and RA types III, VII, and VIII tires. Tires of similar proportions shall also be classified as one of these three groups for growth determination.

3. Growth determination shall be as follows, using factors from table I:

$$W_G = G_W.W$$

$$D_G = D + G_H.H$$

$$W_{SG} = G_W.W_S$$

$$D_{SG} = D + G_H.H_S$$

$$H = \frac{D_O - D}{2}$$

$$H_S = \frac{D_S - D}{2}$$



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.

(d) The chart on figure 2 does not cover twin tires or tires in tandem, and clearance allowances between tires should be determined as follows:

(1) Twin tires - The distance between the CL of tires shall be 1.18 $W_G$.

(2) Tandem tires - The distance between the axle CL of forward and rear tires shall be:

$D_G$ + 0.20 $W_G$ for tires 10 inches in width and less.
$D_G$ + 0.15 $W_G$ for tires above 10 inches in width.

## 4. INTERNATIONAL STANDARDIZATION

*4.1 International standardization agreement. Certain provisions of this standard are the subject of international standardization agreement (ASCC Air Standard 17/14). When amendment, revision, or cancellation of this standard is proposed, which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

(The margins of this standard are marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.)

Custodians:
Army - AV
Navy - AS
Air Force - 11

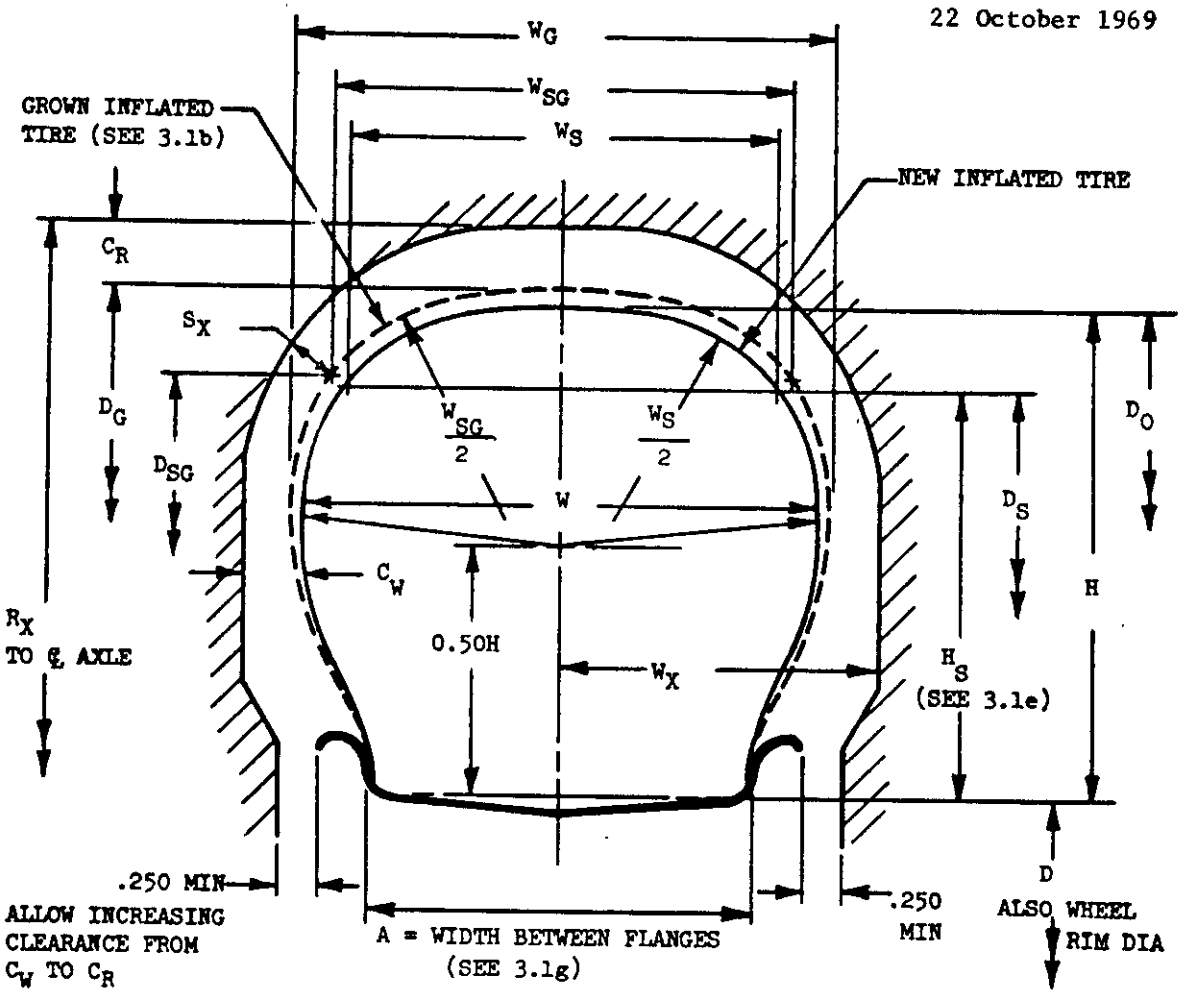
Review activities:
Army - AV
Navy - AS
Air Force - 70

Preparing activity:
Air Force - 11

Project No. 2620-0035



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.




GROWN INFLATED TIRE

$W_G$ = MAXIMUM GROWN SECTION WIDTH

$D_G$ = MAXIMUM GROWN OUTSIDE DIAMETER

$W_{SG}$ = MAXIMUM GROWN SHOULDER WIDTH

$D_{SG}$ = MAXIMUM GROWN SHOULDER DIAMETER

NEW INFLATED TIRE

$D_O$ = MAXIMUM OUTSIDE DIAMETER

H = MAXIMUM SECTION HEIGHT

W = MAXIMUM CROSS SECTION WIDTH

$D_S$ = MAXIMUM SHOULDER DIAMETER

$H_S$ = MAXIMUM SHOULDER HEIGHT

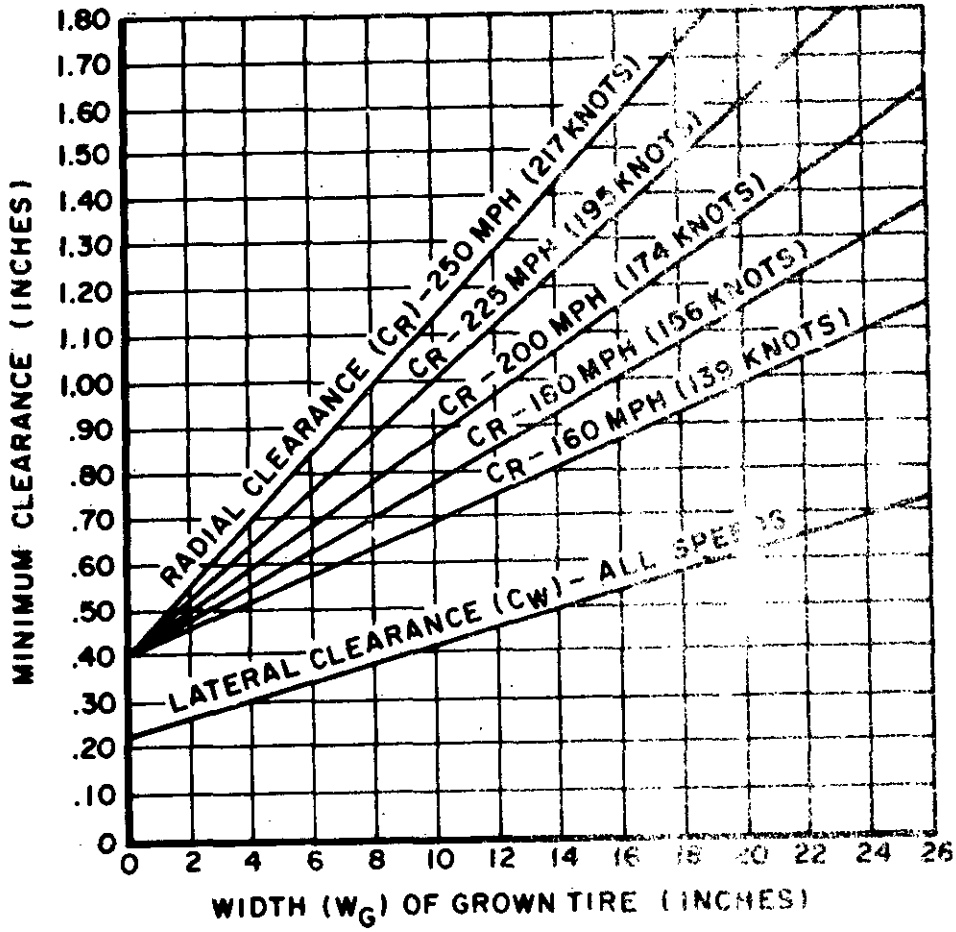
$W_S$ = MAXIMUM SHOULDER WIDTH

A = WIDTH BETWEEN FLANGES

FIGURE 1. Profile Proportions for Aircraft Landing Wheel Tires



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.



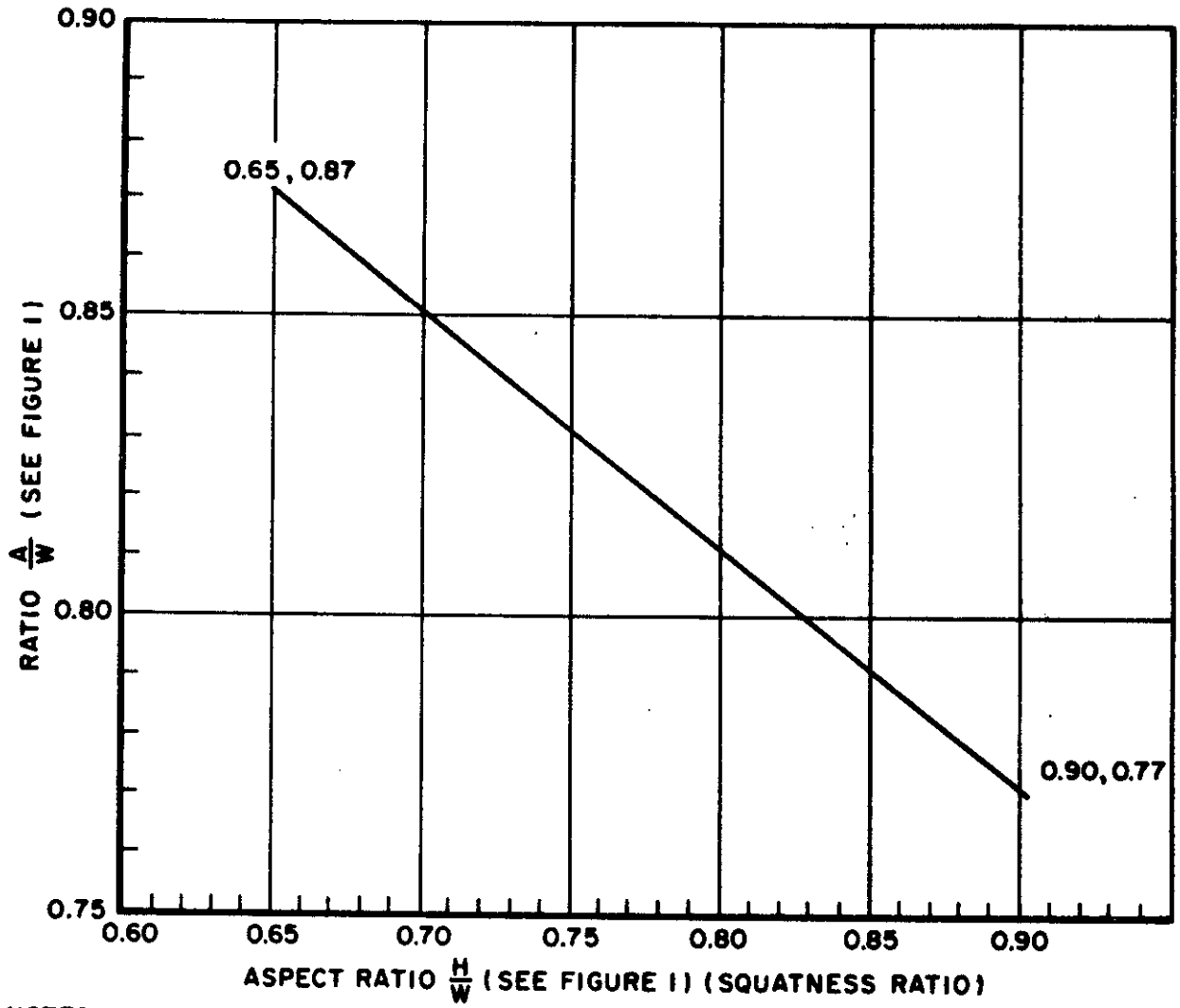

NOTES:

1. MINIMUM CLEARANCE DIAMETER EQUALS MAXIMUM GROWN DIAMETER PLUS ($2 \times C_R$).
2. MINIMUM CLEARANCE WIDTH EQUALS MAXIMUM GROWN WIDTH PLUS ($2 \times C_W$).
3. MINIMUM CLEARANCE AT SHOULDER REGION EQUALS $\frac{C_R + C_W}{2}$.

*FIGURE 2. Chart for Obtaining Radial Clearance $C_R$ and Lateral Clearance $C_W$



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.




NOTES:

1. THE MEAN OR NOMINAL DIMENSIONS H AND W PERTAIN TO A NEW UNUSED INFLATED TIRE.
2. WIDTH BETWEEN FLANGES (DIMENSION A ON FIGURE 1) TO BE ADJUSTED TO NEAREST 1/4 INCH.
3. EQUATION FOR LINE FROM 0.65 TO 0.90 ASPECT RATIO "Y" EQUALS $1.13-\frac{X}{2.5}$.

FIGURE 3. Design Guide for Obtaining Dimension "A"



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

ASD/ENYES
Wright-Patterson AFB, OH 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:01Z
Check the source to verify that this is the current version before use.